

Meeting Summaries of July 2

Date: July 2, 2026

Time: 1:30 - 3:30 Vancouver

Place: TRU, S203

Attendees: Feng, Prof. Javed, Prof. Yue

Topic: In-person meeting. Feng reports the next proposition of his work after last meeting; questioning and feedback from the advisors; discussion on improvements and next steps.

Checklist of Accomplishments from Last Meeting

- Feng has exchanged emails with Jian and prof. Jan for checking the previous chemical data values and units. Jian helped Feng correct an unit-usage issue in elements Al, Fe and Ca, they should be in mg/g instead of ug/g. It improved analysis quality but did not fully resolve the complete-duplication issue.
- Prof. Jan replied that he is working on examination of the chemical data, and he will send back a validated chemical data set later.
- Feng has fixed some issues in slides page-9, the $Var(\cdot)$ formula and the notations around matrix, more readable and understandable.

Questions, Feedbacks and Suggestions

- It needs to untangle the confusing transformation and standardization steps in the PCA computation process. Several interactions between the operations need to be sorted out:
 1. Where does the unit-effect come from and exist in the processing chain? When or in which step does it get removed?
 2. How is the log-transformation interacting with the unit-effect? Should the unit-effect be removed before log-transformation or after log-transformation? Or, is the log-transformation itself a way of removing the unit-effect?
 3. How is the standardization(z-score way) interacting with the unit-effect? Then where should we put the standardization step in the processing chain and how does its positioning affect the unit-effect and interaction with log-transformation?
 4. The above value-processing steps are not only for computing loadings, $\mathbf{L}$, in PCA, but exist in computing the principal component values (stressor values) with ($\mathbf{M}$ and $\mathbf{L}$).
- In arguing which PC loadings should be remained or removed, it needs citation to support it but not just common sense.

- On page-13, the summation formula of adding toxic principal component values needs to adjust the PC subscripts to account for the removed PC loadings. Or, Feng needs to rephrase the expression of the loading remaining operation so that a simpler notation language can be used for the summation formula.
- To show how the integrated contamination score vary among sites, such as site i -th has additive-view contamination score as $SumRel_i = \sum_j \mathbf{S}_{i,j} = \sum_j (\mathbf{M}'\mathbf{L})_{i,j}$, it is clearer to use histogram to show the distribution of $SumRel_i$ over sites.

Action Items

- Feng needs to fix these mentioned issues, bring the fixes and prepare the next propotion for the next meeting.

Summary of Discussion

- Feng reported how the chemical data was processed in the PCA computation from original chemical matrix $\mathbf{M}$ to processed matrix $\mathbf{M}'$ and then to the PCA loadings matrix $\mathbf{L}$ and the stressor values matrix $\mathbf{S} = \mathbf{M}'\mathbf{L}$.
- Feng gave an introduction about extended work of why and how to defence and adjust a proper cut-off value for choosing least-polluted sites as natural benchmark sites where ecological communities not-affected by contamination.

Next Meeting Time

- **Date and Time:** July 7, 2026, 1:30 - 3:30 Vancouver
- **Form:** In-person. Science building, room S203, TRU.